2021-10585 설현웅 13주차

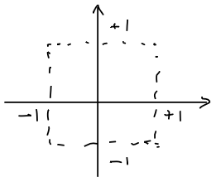
1번. construct set C in plane

that is ① convex

② symmetric about 0

③ area 4

④ contains no nonzero integral point.



$C = \{(x,y) \in \mathbb{R}^2 : |x|<1,\ |y|<1\}$

$= (-1,1) \times (-1,1)$

- 넓이 $= \overset{가로}{\overbrace{2}} \cdot \overset{세로}{\overbrace{2}} = 4$

- $(x,y) \in C$ then $|x|, |-x|, |y|, |-y| < 1$ (원점 대칭)

  $(-x,-y) \in C$

- 정수점 in $C \iff (x,y) \in \mathbb{R}^2,\ -1 < x, y < 1 \iff x = y = 0$ (원점)

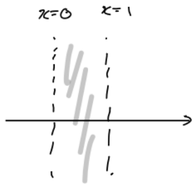
- 정사각형 내부의 두 점 사이를 이으면 $\Rightarrow$ convex

  선분이 정사각형 안에 머무름

2번. set C in plane ① convex

② infinite area

③ no integral point.



$C = \{(x,y) \in \mathbb{R}^2 : 0 < x < 1\}$

- x 좌표가 $0, 1$ 사이 $\Rightarrow$ 선분 위 x 좌표도 0과 1 사이

- $\text{area}(C) =$ 밑변 $\times$ 높이 $= \infty$

  $1 \times \infty$

- $(m,n) \in \mathbb{Z}^2 \iff 0 < m < 1$ $\rightarrow\leftarrow$

6번. if ① $a \not\equiv 0 \pmod p$ $\Rightarrow$ $\exists\, x, y \in \mathbb{Z}$

② p prime

③ any positive integer n

s.t ① $x \equiv ay \pmod p$

② $0 < x \le n$

③ $|y| \le \frac{p}{n}$

- $\Lambda = \{(x,y) \in \mathbb{Z}^2 : x \equiv ay \pmod p\}$

  $= \{m(p,0) + n(a,1) : m, n \in \mathbb{Z}\}$

  $\because$ $x - ay = pk$ ($\exists k$)

  $\Rightarrow x = pk + ay$

  $y = y$

  $\Rightarrow k \cdot (p,0) + y(a,1)$

- $\det \Lambda = \det \begin{bmatrix} p & 0 \\ a & 1 \end{bmatrix} = p$

  $\because$ $x = mp + na \Rightarrow x \equiv na \pmod p$

  $y = n$

- $C_\varepsilon = \{(x,y) \in \mathbb{R}^2 : |x| < n + \varepsilon,\ |y| < \frac{p}{n} + \varepsilon\}$ $(\varepsilon > 0)$

  $\text{area}(C_\varepsilon) = 2(n+\varepsilon) \cdot 2\left(\frac{p}{n} + \varepsilon\right) \underbrace{>}_{} 4p = 4\det(\Lambda)$

  because $\varepsilon > 0$

- by Minkowski C_ε에는 0이 아닌 격자점이 존재한다.

  i.e $\exists (x,y) \ne (0,0)$, $\in \mathbb{Z}^2$ s.t ① $x \equiv ay \pmod p$

  ② $|x| < n + \varepsilon$, $|y| < \frac{p}{n} + \varepsilon$

  with $\varepsilon \to 0$ $|x| \le n$ $|y| \le \frac{p}{n}$

- if $x = 0$ then $0 \equiv ay \pmod p$ $\because$ $a \not\equiv 0 \pmod p$

  $a^{-1} \cdot 0 \equiv a^{-1} a y$ (··)

  $0 \equiv y$ (··)

  however $|y| \le \frac{p}{n}$, $|y| < p$ if $n \ge 2$ $\Rightarrow y = 0$ $\rightarrow\leftarrow$

- $x \equiv ay \pmod p$ then $-x \equiv a(-y) \pmod p$ 따라서 $(-x,-y)$를 사용.

- $n = 1$ then $a\underset{y}{\underset{\shortparallel}{a^{-1}}} \equiv \underset{x}{\underset{\shortparallel}{1}} \pmod p$ then $0 < x \le 1$, $|y| \le p$.

14. any lattice Λ in plane

contains nonzero (x, y) s.t $|xy| \le \frac{1}{2} d(\Lambda)$

- by Minkowski

: Λ에 대하여 ① 원점 대칭

② convex 인 C of area$(C) > 4d(\Lambda)$

$\Rightarrow$ C에는 nonzero 격자점이 있다.

- $C_\varepsilon = \{(x, y) \in \mathbb{R}^2 : \frac{|x|}{A} + \frac{|y|}{B_\varepsilon} < 1\}$ (마름모)

with ① $A > 0$

② $AB_\varepsilon = 2d(\Lambda) + \varepsilon$

$\longrightarrow \begin{cases} \text{원점 대칭} \\ \text{convex} \end{cases}$

$$\text{넓이} = 2AB_\varepsilon = 2(2d(\Lambda) + \varepsilon) = 4d(\Lambda) + 2\varepsilon$$

Minkowski!

- C_ε에는 nonzero 격자점 있음 .

i.e $\frac{|x|}{A} + \frac{|y|}{B_\varepsilon} < 1$ and $(x, y) \ne (0, 0)$

- $u = \frac{|x|}{A}$, $v = \frac{|y|}{B_c}$ $u + v < 1$ $\Rightarrow$ $\boxed{uv} \le \left(\frac{u+v}{2}\right)^2 < \frac{1}{4}$

$$uv = \frac{|xy|}{AB_\varepsilon}$$

$\Rightarrow$ $|xy| < AB_\varepsilon / 4 = \frac{1}{2} d(\Lambda) + \frac{\varepsilon}{4}$ $\Rightarrow$ $\varepsilon \to 0$

$|xy| \le \frac{1}{2} d(\Lambda)$

- $|xy| = \frac{1}{2} d(\Lambda)$ then $\frac{|x|}{A} = \frac{|y|}{B} = \frac{1}{2}$ $\Longrightarrow$ $|x| \ne \frac{A}{2}$가 되도록 A를 선택 가능

$|x| = \frac{A}{2}$

lattice generator A that can replace

the matrix used in Step 1 of pf of Lagrange Four Square thm.

while still satisfying necessary cond for proof.

- choose $A' = \begin{bmatrix} p & 0 & v & s \\ & p & -s & v \\ & & 1 & \\ & & & 1 \end{bmatrix}$

- choose integers s.t $v^2 + s^2 \equiv -1 \pmod{p}$

- lattice $\Lambda = A'\mathbb{Z}^4$

- 임의의 $\vec{x} \in \Lambda$에 대해 $\vec{t} = (t_1, t_2, t_3, t_4)$가 존재.

$$\Rightarrow \vec{x} = A'\vec{t} = \begin{bmatrix} pt_1 + vt_3 + st_4 \\ pt_2 - st_3 + vt_4 \\ t_3 \\ t_4 \end{bmatrix}$$

- $x_1^2 + x_2^2 + x_3^2 + x_4^2 = (pt_1 + vt_3 + st_4)^2 + (pt_2 - st_3 + vt_4)^2 + t_3^2 + t_4^2$

$$\pmod{p} \equiv v^2t_3^2 + 2vst_3t_4 + s^2t_4^2 + s^2t_3^2 - 2vst_3t_4 + v^2t_4^2 + t_3^2 + t_4^2$$

$$\equiv (v^2 + s^2 + 1)(t_3^2 + t_4^2)$$

$$\equiv 0$$

$\det A' = p \cdot p \cdot 1 \cdot 1 = p^2$

(위삼각행렬)

① $\forall \vec{x} \in A'\mathbb{Z}^4$

$x_1^2 + x_2^2 + x_3^2 + x_4^2 \equiv 0 \pmod{p}$

② $\det A' = p^2$

3번. infinite conti fractions

(a) $\langle 2, 2, 2, 2, \ldots \rangle$

$x = 2 + \frac{1}{x}$ ⇒ since $x > 0$

$x^2 = 2x + 1$ $\quad x = 1 + \sqrt{2}$

$x^2 - 2x - 1 = 0$

$x = 1 \pm \sqrt{2}$

(b) $\langle 1, 2, 1, 2, 1, 2 \ldots \rangle$

$x = 1 + \dfrac{1}{2 + \frac{1}{x}}$ $\quad x > 0$

therefore $\dfrac{1+\sqrt{3}}{2} = x$

$x - 1 = \dfrac{x}{2x+1}$

$(x-1)(2x+1) = x$

$x = \dfrac{1 \pm \sqrt{3}}{2}$

(c) $\langle 2, 1, 2, 1 \ldots \rangle$

$x = 2 + \dfrac{1}{1 + \frac{1}{x}}$ $\quad x > 0$

therefore

$(x-2) = \dfrac{x}{(x+1)}$ $\quad x = 1 + \sqrt{3}$

$(x-2)(x+1) = x$

$x = 1 \pm \sqrt{3}$

(d) $\langle 1, 3, \underbrace{1, 2, 1, 2 \ldots}_{''\, y} \rangle$

$x = 1 + \dfrac{1}{3 + \frac{1}{y}}$

$= 1 + \dfrac{y}{3y+1}$ $\quad$ with $y = \dfrac{1+\sqrt{3}}{2}$

$= 1 + \dfrac{1+\sqrt{3}}{5 + 3\sqrt{3}}$

$= 1 + (2 - \sqrt{3}) = 3 - \sqrt{3}$

4번. $(n \geq 1)$ $\quad k_n / k_{n-1} = \langle a_n, a_{n-1}, \ldots, a_2, a_1 \rangle$

similar contined fraction expression for h_n/h_{n-1} $(a_0 > 0)$

Γ.:

$h_n = a_n h_{n-1} + h_{n-2}$ $\quad (h_{-2}, k_{-2}) = (0, 1)$

$k_n = a_n k_{n-1} + k_{n-2}$ $\quad (h_{-1}, k_{-1}) = (1, 0)$

$(h_0, k_0) = (a_0, 1)$

- $k_n = a_n k_{n-1} + k_{n-2}$

$$\frac{k_n}{k_{n-1}} = a_n + \frac{1}{k_{n-1}/k_{n-2}} = a_n + \frac{1}{a_{n-1} + \frac{1}{k_{n-2}/k_{n-3}}} = \cdots = a_n + \frac{1}{a_{n-1} + \frac{1}{\cdots + \frac{1}{a_1}}}$$

$\langle a_n, a_{n-1}, \ldots, a_2, a_1 \rangle$

$''$

- $h_n = a_n h_{n-1} + h_{n-2}$.

with $\dfrac{h_1}{h_0} = \dfrac{a_1 a_0 + 1}{a_0} = a_1 + \dfrac{1}{a_0}$

$$\frac{h_n}{h_{n-1}} = a_n + \frac{1}{h_{n-1}/h_{n-2}} = a_n + \frac{1}{a_{n-1} + \frac{1}{\cdots + \cancel{\frac{h_1}{h_0}}}}$$

$a_1 + \frac{1}{a_0}$ $= \langle a_n, a_{n-1}, \ldots, a_1, \boxed{a_0} \rangle$

5번. $u_0 / u_1 = \langle a_0, a_1, \ldots, a_n \rangle$

: rational number in lower terms

⇒ if $0 \leq i < n$ then $\left| r_i - \frac{u_0}{u_1} \right| \leq \dfrac{1}{k_i k_{i+1}}$

equality iff $i = n-1$

Γ.:

- $\dfrac{u_0}{u_1} = r_n = \dfrac{h_n}{k_n}$

- $h_i k_{i-1} - h_{i-1} k_i = (-1)^{i-1}$

$|h_i k_{i-1} - h_{i-1} k_i| = 1$

- $r_i = h_i / k_i$
  $r_n = h_n / k_n$ ⇒ $|r_i - r_n| = \left| \dfrac{h_i}{k_i} - \dfrac{h_n}{k_n} \right| = \dfrac{|h_i k_n - h_n k_i|}{k_i k_n} \leq \dfrac{1}{k_i k_n}$

$\leq \dfrac{1}{k_i k_{i+1}}$

⇒ $\left| r_i - \dfrac{u_0}{u_1} \right| \leq \dfrac{1}{k_i k_{i+1}}$

- 등호 성립 ⇒ $k_n = k_{i+1}$

with $k_j = \boxed{a_j}\, k_{j-1} + k_{j-2}$ 이므로 $n = i+1$ (뒤에 항 無)

≥ 1

- $i = n-1$ ⇒ $r_{n-1} = r_i$ ⇒ $\left| r_i - \frac{u_0}{u_1} \right| = \dfrac{1}{k_i k_{i+1}}$

$r_n = u_0 / u_1$

$|r_{n-1} - r_n| = \dfrac{1}{k_{n-1} k_n}$

증명. ① prime $p \equiv 1 \pmod 4$

② integer $u^2 \equiv -1 \pmod p$

③ $\frac{u}{p} = \langle a_0, a_1, \dots, a_n \rangle$

$\frac{h_j}{k_j} = \langle a_0, a_1, \dots, a_j \rangle$

④ i는 $k_i \le \sqrt{p}$ 인 가장 큰 정수

Claim $i < n$

- $h_n / k_n = u/p$ but $p > \sqrt{p}$ $\Rightarrow$ $k_i \le \sqrt{p} < \boxed{k_{i+1}}$
- $k_n = p$

$k_n = p > \sqrt{p} \ge k_i$

$n > i$

Claim $\left| \frac{h_i}{k_i} - \frac{u}{p} \right| < \frac{1}{k_i \sqrt{p}}$, $|h_i p - u k_i| < \sqrt{p}$

유리수 $u/p = \langle a_0, a_1, \dots, a_n \rangle$

$\Rightarrow$ $0 \le j < n$ then $\left| \frac{h_j}{k_j} - \frac{u}{p} \right| \le \frac{1}{k_j k_{j+1}}$ $\Rightarrow$ $\left| \frac{h_i p - u k_i}{k_i p} \right| < \frac{1}{k_i \sqrt{p}}$

$\boxed{\left| \frac{h_i}{k_i} - \frac{u}{p} \right| \le \frac{1}{k_i k_{i+1}} < \frac{1}{k_i \sqrt{p}}}$ $|h_i p - u k_i| < \frac{k_i p}{k_i \sqrt{p}}$

$= \sqrt{p}$

Claim $0 < x^2 + y^2 < 2p$

let $(x, y) = (\underbrace{\boxed{k_i}}_{\le \sqrt{p}}, \underbrace{\boxed{h_i p - u k_i}}_{< \sqrt{p}})$ $\Rightarrow$ $x^2 \le p$ $\Rightarrow$ $x^2 + y^2 < 2p$

$y^2 < p$

meanwhile $x = k_i > 0$ therefore $x^2 + y^2 > 0$

$0 < x^2 + y^2 < 2p$

$\downarrow$

$x^2 + y^2 = p$

Claim $x^2 + y^2 \equiv 0 \pmod p$

$x = k_i$ $\Rightarrow$ $y^2 \equiv u^2 x^2$

$y = h_i p - u k_i \equiv -u k_i \pmod p$

$\equiv -ux$

$x^2 + y^2 \equiv (1 + u^2) x^2$

$\equiv (1 - 1) x^2 \equiv 0$

two distinct inf simple continued fractions converge to diff values

$\alpha = \langle a_0, a_1, \dots \rangle$ $a_m \ne b_m$

$\beta = \langle b_0, b_1, \dots \rangle$

if $\alpha = \beta$ $\alpha = a_0 + \boxed{\frac{1}{\langle a_1, a_2 \dots \rangle}}$ then $\langle a_1, a_2, \dots \rangle > 1$

between 0 and 1

$\Rightarrow$ $a_0 < \alpha < a_0 + 1$ $\Rightarrow$ $\boxed{a_0 = b_0}$

$a_0 = \lfloor \alpha \rfloor = \lfloor \beta \rfloor = b_0$

$\frac{1}{\langle a_1, a_2, \dots \rangle} = \alpha - a_0 = \beta - b_0 = \frac{1}{\langle b_1, b_2 \dots \rangle}$

$\langle a_1, a_2 \dots \rangle = \langle b_1, b_2 \dots \rangle$ $\Rightarrow$ $a_1 = b_1$

$\Rightarrow$ $a_2 = b_2$

....